

SEDAR 55 South Atlantic Vermilion Snapper

- Further investigate discrepancies between age composition data and indices of abundance
- Further develop methods to standardize and combine SERFS chevron trap and video gears for creating indices of abundance
- Evaluate sample size cutoffs and weighting procedures for age and length compositions. What should be the minimum standards, and how does this interplay with the number of age and length classes modeled in the assessment?
- In stock assessment, various likelihood formulations have been used for fitting age and length composition data. The multinomial distribution and its robust versions have been the most widely applied. However, more recently the Dirichlet-multinomial and logistic-normal have attracted attention. A simulation study could shed light on the performance of these various likelihood formulations under sampling conditions realistic in the southeast U.S.
- Vermilion snapper were modeled in this assessment as a unit stock off the southeastern U.S. For any stock, variation in exploitation and life-history characteristics might be expected at finer geographic scales. Modeling such sub-stock structure would require more data, such as information on the movements and migrations of adults and juveniles, as well as spatial patterns of larval dispersal and recruitment. Even when fine-scale spatial structure exists, incorporating it into a model may or may not lead to better assessment results (e.g., greater precision, less bias). Spatial structure in a vermilion snapper assessment model might range from the very broad (e.g., a single Atlantic stock) to the very narrow (e.g., a connected network of meta-populations living on individual reefs). What is the optimal level of spatial structure to model in an assessment of snapper-grouper species such as vermilion snapper? Are there well defined zoogeographic breaks (e.g., Florida keys, Cape Hatteras) that should define stock structure? How much connectivity exists between the Gulf of Mexico and Atlantic stocks?